Geometric and Performance Analysis of Recursive Multi-Agent Systems for Enterprise Process Automation

Problem Statement

Enterprise processes such as procurement, order fulfilment, customer support, supply-chain coordination, and IT service management involve interconnected tasks requiring planning, information retrieval, tool invocation, decision-making, and verification. LLM-based multi-agent systems can automate such workflows, but their reliability may degrade as the number of process steps, dependencies, tools, and agent interactions increases. Inspired by *Recursive Multi-Agent Systems (RecursiveMAS)*, this project will develop a recursive multi-agent architecture for enterprise workflows and evaluate it from both conventional and geometric perspectives. In addition to task success, tool-calling accuracy, latency, token usage, and computational cost, the project will analyze the geometry of agent representations to understand how latent states evolve with workflow complexity, recursion depth, and successful or failed execution.

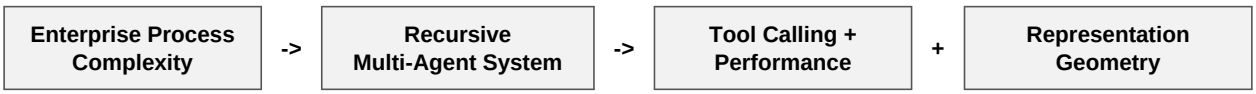
Objectives

1. Develop a Recursive Multi-Agent System for Enterprise Processes. Design and implement a workflow with specialized agents such as **Planner, Tool-Caller/Executor, and Verifier**. Create enterprise workflows of increasing complexity by varying process steps, branches, dependencies, tools, and recursion rounds.

2. Evaluate Agent and Tool-Calling Performance. Study how workflow complexity and recursion depth affect **task success rate, tool-selection accuracy, tool-argument accuracy, tool-call sequence correctness, latency, token consumption, number of tool calls, and computational cost**. Compare recursive and non-recursive/text-mediated baselines where feasible.

3. Perform Geometric Analysis of Agent Representations. Extract latent representations at different workflow stages and recursion rounds. Characterize them using **PCA/SVD, effective rank, cosine similarity, representation drift, inter-agent distances, and subspace/principal-angle similarity**. Compare successful and failed executions to determine whether geometric changes correlate with tool-calling errors and performance degradation.

Experimental Framework



Core Research Question: Do geometric properties of latent representations provide indicators of reliability and failure in recursive multi-agent systems as enterprise process complexity increases?

Suggested application: Enterprise procurement/order processing - inventory lookup, vendor selection, price comparison, approval, purchase-order generation, and delivery tracking - with controlled easy, medium, and complex workflow variants.

Basis: Zou et al., *Recursive Multi-Agent Systems* (2026). The project extends its recursive latent collaboration, tool-integrated deliberation, accuracy/runtime/token evaluation, and PCA-based semantic analysis toward enterprise workflows and quantitative geometric representation analysis.